

THE NONCONVEX A_3 COXETER-CHAMBER ATLAS: CONGRUENCE RIGIDITY, EXACT COVERS, AND PATTERSON SEPARATION

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ABSTRACT. The barycentric A_3 dissection of a regular tetrahedron consists of 24 orthoscheme chambers naturally indexed by S_4 . Earlier work classified the 15 connected convex Coxeter-pure tiling orbits, reported an exhaustive audit count of 82 connected nonconvex left-translate tiling orbits, and posed the classification of nonconvex congruent tilers as an open problem. We close that problem within the fixed Coxeter-pure carrier. On all 2,874 eligible connected chamber-union orbits, an exact rank-three corner-locus audit proves that free Euclidean congruence is precisely left S_4 equivalence. Hence congruent Coxeter-pure covers are exactly left-translate covers. We then construct the complete canonical atlas behind the earlier count and determine its cover, contact, and symmetry structure.

The 82 nonconvex orbits have chamber sizes 6, 8, 12, with multiplicities 14, 18, 50. The element-labelled right Patterson function $a_U(h) = |U \cap Uh|$ separates all atlas rows, whereas three natural compressions yield only 20, 33, and 32 classes. We enumerate 786 exact covers, 86 cover orbits, 1,148 multiplier selectors, and 102 selector orbits; every selector-independent carrier is a typed double coset. Rank-coloured contact complexes distinguish information invisible to their rank marginals and produce two certified local-to-global obstruction pairs. The atlas has 82 full and 159 oriented tile-congruence classes; its 86 partition classes split into 151 oriented classes, all are tile-transitive inside the tetrahedron, and 69 are rotation-tile-transitive. Every result is an exact finite statement for the fixed A_3 carrier: every partition member is a union of its chambers. The previously published count $15 + 82$ is prior art and a replay target, not a novelty claim.

Keywords. Coxeter complex; regular tetrahedron; congruent dissection; exact cover; symmetric group; Patterson function; homometry; coloured flag complex.

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1. INTRODUCTION

1.1. From the convex catalogue to the nonconvex atlas. The barycentric subdivision of a regular tetrahedron is simultaneously a geometric dissection into 24 congruent orthoschemes and the spherical Coxeter complex of type A_3 coned to the barycentre. This coincidence makes a finite classification possible: unions of chambers become subsets of S_4 , adjacency is a Cayley-graph relation, and tetrahedral relabelling is left multiplication. The finite model is small enough for exhaustive enumeration but large enough for nonconvexity, multiple exact covers, chirality, and contact phenomena that do not occur in the convex catalogue.

The convex theory was developed in [Bab26]. There, connected convex chamber unions were identified with poset cones, and the tiling pieces were classified by the 15 series-parallel poset types on four elements. The same project reported an independent exhaustive count

$$97 = 15 + 82$$

of connected left-translate tiling orbits, with 82 nonconvex rows, and posed the classification of connected nonconvex chamber unions that tile by congruent copies as an open problem. The numerical split is therefore self-prior-art. The purpose of this paper is to close the congruence gap left by that audit and replace the bare count by a canonical, structurally complete, and independently replayable atlas.

Four questions drive the construction.

- (i) Does free Euclidean congruence enlarge a left S_4 orbit anywhere in the complete eligible chamber-union carrier?
- (ii) Can the 82 rows be separated by an intrinsic invariant that respects the locked tetrahedral action?
- (iii) How many inequivalent exact covers does each tile admit, and what is the algebraic structure of its multiplier choices?
- (iv) Which contact and symmetry data survive after local compression, and where do local equivalences fail to assemble globally?

The answers use four different layers. An all-candidate rank-three corner-locus theorem identifies free congruence with the locked left action. The labelled right Patterson function separates tile orbits. Exhaustive exact-cover and multiplier enumeration describes all partitions inside the finite carrier. Exact rank-coloured barycentric intersections then recover contact, handedness, and finite-container transitivity.

1.2. Main results. The main theorem packages the numerical statements; precise definitions and stronger typed formulations appear in later sections.

Theorem 1.1 (Nonconvex A_3 chamber atlas: overview). *In the fixed 24-chamber A_3 dissection of the regular tetrahedron, under the action and equivalences of Section 2, the following hold.*

- (a) *There are exactly 82 free-congruence classes of connected halfspace-nonconvex pieces that tile by congruent Coxeter-pure copies. Every such copy is a left S_4 translate. Their chamber sizes are 6, 8, 12, with multiplicities 14, 18, 50.*

- (b) *The full element-labelled right Patterson vector separates the 82 orbits. Conjugacy-class sums, within-class coordinate multisets, and the fully unlabelled coordinate multiset yield respectively 20, 33, 32 classes.*
- (c) *The 82 tiles admit 786 exact covers in 86 global-left cover orbits. Their 1,148 multiplier selectors form 102 selector orbits, with the exact taxonomy $90 + 2 + 10$ described in Section 5.*
- (d) *Canonical representatives of the 86 cover orbits have 212 within-cover contact edges and 82 edgewise signatures. Two pairs are edgewise equivalent but not globally aligned.*
- (e) *The atlas has 82 full Euclidean and 159 orientation-preserving tile classes, split into 77 chiral and 5 achiral full classes. The 86 partition classes split into 151 oriented classes; all 86 are finite-container tile-transitive and 69 are rotation-tile-transitive.*

Part (a) combines the left-exact-cover census with the independent all-candidate rigidity theorem Theorem 7.3. The latter ranges over all 2,874 eligible connected chamber-union orbits, not merely the 82 positive rows, and shows that every congruent Coxeter-pure cover is already a left-translate cover; see Corollary 7.4. Here *Coxeter-pure* means that every partition member remains a union of the fixed A_3 chambers.

The result in part (b) is deliberately bounded. Patterson functions and homometry are classical [MGA⁺11, RS82, GP18]. We prove injectivity only on the displayed 82-row family, not for arbitrary subsets of a finite group. Likewise, the contact and symmetry statements are about a finite tetrahedral container, not a periodic monohedral tiling of Euclidean space.

1.3. Why the labels and types matter. Several superficially similar objects must remain distinct. A tile orbit is not a cover orbit; a cover is not a multiplier selector; the stabiliser of a tile is not the stabiliser of a cover; an edgewise family of contact equivalences is not one globally coherent equivalence. Losing these types creates false uniqueness and false local-to-global statements. The paper therefore fixes all actions, quotient relations, and quantifier orders before stating the enumerative results.

The same discipline explains the Patterson theorem. Left multiplication fixes every labelled coordinate, while right multiplication conjugates the coordinate labels. The values alone do not suffice: all three natural label compressions collide. Thus the nonabelian labelling is not decorative data; it is the separating information on this carrier.

1.4. Logical status and organisation. The three-way equivalence between halfspace, chamber-geodesic, and Euclidean convexity for connected A_3 chamber unions is imported from [Bab26]. All enumerations, Patterson identities, exact-cover classifications, contact comparisons, and ambient congruence calculations in this paper use exact integer or rational arithmetic. The companion package fully reconstructs the connected and eligible census, the global congruence gate, and the congruent-copy exact-cover closure. The Patterson, cover, selector, contact, boundary, and ambient-symmetry layers are distributed as exact parent-pinned snapshots with record hashes and public structural recheckers; their production builders remain internal. In particular, the congruence statement is deduced only after the complete

eligible carrier has passed both exact isometry lanes. A machine certificate supports a finite proof; its stated replay level is part of the claim boundary and does not enlarge the theorem's scope.

Section 2 fixes the geometric and group-theoretic model. Section 3 proves the finite reduction and constructs the atlas. Section 4 develops the separating invariant. Sections 5 and 6 classify covers, selectors, and contact complexes. Section 7 treats congruence and transitivity, and Section 8 records a presentation-sensitivity result. The final sections delimit prior art and give the replay contract. The appendices provide the search proof, the complete 82-row catalogue, and a theorem-to-certificate crosswalk.

2. THE FIXED A_3/S_4 MODEL

2.1. The tetrahedron and its Coxeter chambers. Let

$$H = \left\{ x \in \mathbb{R}^4 : \sum_{i=0}^3 x_i = 1 \right\}, \quad T = \text{conv}\{e_0, e_1, e_2, e_3\} \subset H,$$

with the induced Euclidean metric. This is a regular tetrahedron of edge length $\sqrt{2}$; rescaling does not affect any statement below. For a nonempty set $I \subseteq \{0, 1, 2, 3\}$ put

$$b_I = \frac{1}{|I|} \sum_{i \in I} e_i.$$

If $\sigma = \sigma_0 \sigma_1 \sigma_2 \sigma_3$ is a one-line permutation, the closed barycentric chamber indexed by σ is

$$\Delta_\sigma = \text{conv} \left(b_{\{\sigma_0\}}, b_{\{\sigma_0, \sigma_1\}}, b_{\{\sigma_0, \sigma_1, \sigma_2\}}, b_{\{0, 1, 2, 3\}} \right).$$

The 24 chambers have disjoint interiors and union T .

Write $G = S_4$. We compose permutations as functions, $(pq)(i) = p(q(i))$. The coordinate permutation

$$\Phi_g(x)_i = x_{g^{-1}(i)}$$

is an isometry of T and satisfies $\Phi_g(\Delta_\sigma) = \Delta_{g\sigma}$. Hence the full isometry group of T is realised by left multiplication on chamber labels; its orientation-preserving subgroup is A_4 .

Let s_i exchange positions i and $i + 1$ for $i = 0, 1, 2$. Two chambers share a facet precisely when their labels differ by right multiplication by one s_i . The chamber graph is therefore

$$\Gamma = \text{Cay}_{\text{right}}(S_4, \{s_0, s_1, s_2\}).$$

Left multiplication is a graph automorphism because $g(\sigma s_i) = (g\sigma)s_i$.

2.2. Pieces, masks, and convexity.

Definition 2.1 (Coxeter-pure piece). A piece is a nonempty subset $U \subseteq G$, identified with the closed polyhedron

$$|U|_T = \bigcup_{\sigma \in U} \Delta_\sigma.$$

It is *connected* when the induced chamber graph $\Gamma[U]$ is connected. This is facet-connectedness; contact through only an edge or a vertex does not suffice.

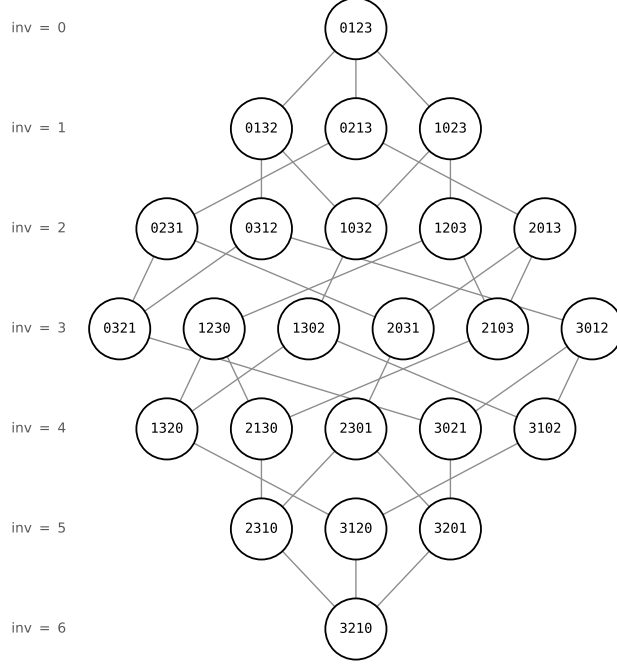


FIGURE 1. The 24 chambers and their facet-adjacency graph. Edges are right multiplication by an adjacent transposition; left S_4 acts by graph automorphisms. Figure reused from [Bab26] under CC BY 4.0.

Order the 24 words lexicographically and let $\text{idx}(\sigma)$ be the zero-based position. The integer mask

$$m(U) = \sum_{\sigma \in U} 2^{\text{idx}(\sigma)}$$

is the primary exact encoding. All masks in the paper are displayed as six hexadecimal digits.

For $i, j \in \{0, 1, 2, 3\}$ define $i <_U j$ when i precedes j in every word of U . These unanimous precedence relations define a poset P_U . Its set of linear extensions is the halfspace closure

$$\text{hcl}(U) = L(P_U).$$

We call U halfspace-convex when $U = \text{hcl}(U)$ and halfspace-nonconvex otherwise. For connected pieces this agrees with chamber-geodesic and Euclidean convexity by the three-way theorem of [Bab26]. The computation uses the halfspace predicate; the geometric terminology invokes that theorem with its stated hypotheses.

2.3. Four typed equivalences.

Definition 2.2 (Tile orbit). The left tile orbit and stabiliser of U are

$$\mathcal{O}_L(U) = \{gU : g \in G\}, \quad H_U = \text{Stab}_L(U) = \{g \in G : gU = U\}.$$

Two pieces are *left-tile equivalent* when they lie in the same $\mathcal{O}_L$ orbit. The canonical representative is

$$\text{can}_L(U) = \arg \min_{V \in \mathcal{O}_L(U)} m(V).$$

The minimum is numeric bitmask order, not lexicographic order on word lists.

Definition 2.3 (Exact cover). An exact left-translate cover of U is an unordered set

$$\mathcal{C} = \{g_1 U, \dots, g_r U\} \subseteq \mathcal{O}_L(U)$$

whose chamber sets are pairwise disjoint and whose union is G . Necessarily $r = 24/|U|$. Covers $\mathcal{C}$ and $\mathcal{C}'$ are *cover equivalent* when $\mathcal{C}' = g\mathcal{C} := \{gV : V \in \mathcal{C}\}$ for one $g \in G$.

Definition 2.4 (Selector). A multiplier selector for a cover $\mathcal{C}$ of U is a set $A \subseteq G$ such that

$$\mathcal{C} = \{aU : a \in A\}.$$

Selectors are not unique when H_U is nontrivial. Their natural global-left action is $A \mapsto gA$.

Definition 2.5 (Free congruence). Pieces U, V are *freely congruent* if an affine Euclidean isometry of H maps $|U|_T$ onto $|V|_T$. This does not, by definition, require the isometry to preserve the ambient tetrahedron. Whole partitions are congruent when one isometry of H maps the complete unordered family of pieces onto the other.

Definition 2.6 (Congruent Coxeter-pure cover). A congruent Coxeter-pure cover is an unordered family $\{U_1, \dots, U_r\}$ of connected pieces in the fixed chamber carrier whose polyhedral interiors are pairwise disjoint, whose union is T , and whose members are pairwise freely congruent. Thus every member is itself a union of chambers of the fixed A_3 dissection. This is the finite-carrier meaning of *by congruent copies* used in the Coxeter-pure problem; copies whose boundaries cut chamber interiors are outside the definition.

The first three definitions are finite group actions. Free congruence is a geometric relation and must be compared with them rather than identified by notation. Right multiplication is also distinct: it conjugates the adjacent-transposition generating set and need not preserve connectivity.

Lemma 2.7 (Basic covariance). *Left multiplication preserves chamber adjacency, cardinality, halfspace closure, exact-cover incidence, and Euclidean shape. Moreover*

$$|\mathcal{O}_L(U)| = \frac{24}{|H_U|}.$$

Proof. Adjacency covariance was observed above. Relabelling the ground set carries the precedence poset P_U to P_{gU} , so it carries their linear extensions and halfspace closures. The other assertions follow from the coordinate isometry Φ_g , and the final identity is orbit-stabiliser. $\square$

3. FINITE REDUCTION AND THE CANONICAL ATLAS

3.1. Candidate space. There are $2^{24} - 1$ nonempty chamber masks. A tiler must be connected and its cardinality must divide 24. We first enumerate connected masks, apply the divisibility filter, and then retain one numeric-minimum representative from each left S_4 orbit. The reduction is summarised in Table 1.

TABLE 1. Exact finite reduction. “Eligible” means connected with chamber cardinality dividing 24.

stage	count
connected nonempty masks	1,210,648
eligible connected masks	67,681
canonical eligible left-orbit representatives	2,874
exact left-translate tilers	97
exact-cover negatives	2,777
convex/nonconvex tiler split	15/82

Lemma 3.1 (Canonical reduction). *Every connected piece is left-equivalent to exactly one stored canonical mask. A canonical piece is a left-translate tiler if and only if the exact-cover recursion below returns a cover.*

Proof. The finite orbit $\{gU : g \in S_4\}$ has a unique minimum integer mask. Left multiplication preserves adjacency by Lemma 2.7. Every cover of any orbit member translates to a cover of the canonical member, and conversely. It therefore remains only to show that the recursion enumerates all covers, which is Lemma 3.3 below. $\square$

3.2. Exact-cover search. For a fixed canonical U , let $\mathcal{O}$ be the set of distinct masks in $\mathcal{O}_L(U)$. A search state is a pair $(R, \mathcal{F})$, where R is the remaining uncovered mask and $\mathcal{F}$ is the chosen family. Initially $R = G$ and $\mathcal{F} = \emptyset$. At a nonterminal state choose the least-indexed chamber c of R and branch over all $V \in \mathcal{O}$ with $c \in V \subseteq R$.

Algorithm 3.2 (Least-uncovered exact-cover recursion). (1) If $R = \emptyset$, output $\mathcal{F}$.
(2) Let c be the least chamber contained in R .
(3) For every $V \in \mathcal{O}$ with $c \in V$ and $V \cap (G \setminus R) = \emptyset$, recurse on $(R \setminus V, \mathcal{F} \cup \{V\})$.

Lemma 3.3 (Exact-cover completeness). *Algorithm 3.2 outputs every unordered exact cover of G by members of $\mathcal{O}_L(U)$ exactly once.*

Proof. Let $\mathcal{C}$ be an exact cover extending the current partial family. The least uncovered chamber c belongs to a unique member V of $\mathcal{C}$. That member is one of the algorithm’s branches, so induction on the number of uncovered chambers reaches $\mathcal{C}$. Conversely, each branch adds a subset of the remaining mask, hence every terminal family is disjoint and covers G . Choosing the least uncovered chamber makes the sequence of chosen members canonical for an unordered cover, so no cover is output twice. $\square$

Two independent implementations support the enumeration. The source-replay lane executes the earlier T2 algorithms at their pinned revision. The independent lane scans all nonempty 24-bit masks, reconstructs the chamber graph, halfspace closure, orbit canonicalisation, and exact-cover recursion without importing the T2 code. The two lanes agree on the entire 97-row positive stream and its exact digest.

3.3. The atlas theorem.

Theorem 3.4 (Canonical nonconvex left-translate atlas). *Up to left S_4 equivalence, exactly 82 connected halfspace-nonconvex pieces tile the reference tetrahedron by left translates. Their distribution is*

$ U $	6	8	12
$\# \text{ orbits}$	14	18	50.

Together with the 15 convex types of [Bab26], they exhaust the 97 positive canonical representatives in Table 1.

Proof. By Lemmas 3.1 and 3.3, the exhaustive candidate scan and exact-cover recursion decide every eligible left orbit. Both implementations find the same 2,777 negatives and 97 positives. The halfspace predicate, calibrated against the imported three-way convexity theorem on connected masks, divides the positives into 15 convex and 82 nonconvex rows. Grouping the latter by popcount gives 14, 18, 50. $\square$

Every atlas row has the content-addressed identifier

$$\text{P18-NC-K } |U| \text{-M } m(U),$$

with the cardinality written in two decimal digits and the mask in six upper-case hexadecimal digits. For example, the first row is P18-NC-K06-M00006F. Source-array positions are locators, never row identity. The complete table appears in Section B.

3.4. Congruence closure of the candidate universe. The left-translate census is only the combinatorial half of the earlier congruent-copy problem. The missing implication is global: a left-exact-cover negative must not become positive by using congruent Coxeter-pure pieces drawn from several left orbits. It would be circular to check this only on the 82 already selected rows.

The ambient argument in Section 7 therefore runs on all 2,874 eligible canonical candidates. Theorem 7.3 proves that free Euclidean congruence equals left S_4 equivalence throughout that carrier, and Corollary 7.4 promotes Theorem 3.4 to the complete classification by congruent Coxeter-pure copies. No equivalence is assumed in the exact-cover search itself.

4. THE ELEMENT-LABELLED PATTERSON SEPARATOR

4.1. Right differences in the group ring. For $U \subseteq G$ write $X_U = \sum_{u \in U} u$ in the integral group ring $\mathbb{Z}[G]$, and let $X_U^{(-1)} = \sum_{u \in U} u^{-1}$. The coefficient of h in $X_U^{(-1)} X_U$ is

$$a_U(h) = |U \cap Uh| = \#\{(u, v) \in U^2 : u^{-1}v = h\}.$$

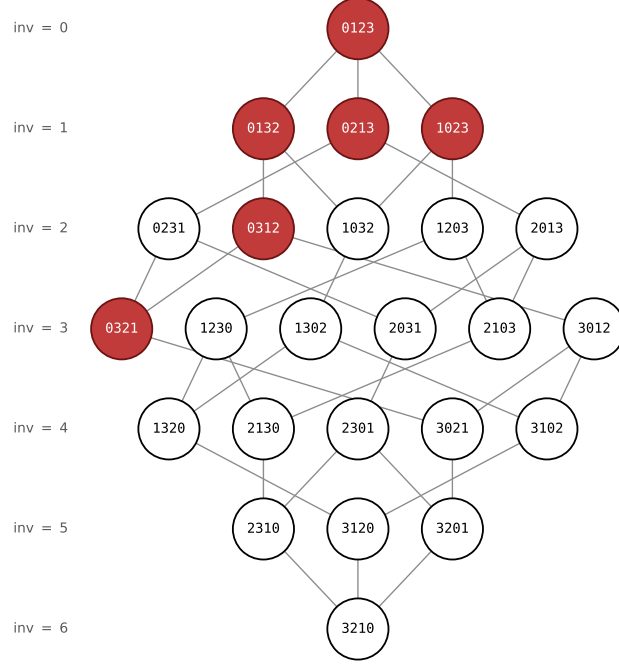


FIGURE 2. The chamber set of the six-chamber row P18-NC-K06-M000077, highlighted in the A_3 chamber graph. This was the existence witness used to show that the nonconvex problem is nonempty in [Bab26]; here it is one row of the complete atlas. Figure reused under CC BY 4.0.

This is the right interval content, or right Patterson function, in counting measure. The left/right distinction is essential in a nonabelian group [MGA⁺11, GP18].

Lemma 4.1 (Patterson identities). *For $U \subseteq G$ and $g, h \in G$,*

$$a_U(e) = |U|, \quad \sum_{h \in G} a_U(h) = |U|^2, \quad a_U(h) = a_U(h^{-1}),$$

$$a_{gU}(h) = a_U(h), \quad a_{Ug}(h) = a_U(ghg^{-1}), \quad a_{G \setminus U}(h) = |G| - 2|U| + a_U(h).$$

Proof. The first identity counts diagonal pairs, the second all ordered pairs in U^2 , and inversion exchanges (u, v) with (v, u) . In a left translate, $(gu)^{-1}(gv) = u^{-1}v$. In a right translate, $(ug)^{-1}(vg) = g^{-1}u^{-1}vg$, which gives the displayed conjugation after renaming the argument. Finally, expand the two complement indicators in $\sum_x (1 - 1_U(x))(1 - 1_U(xh^{-1}))$. $\square$

The formula for right translation explains why coordinate labels cannot be discarded casually. Left translation fixes the vector pointwise; right translation permutes it through inner automorphisms.

4.2. Three controlled compressions. Let $C_1, \dots, C_5$ be the conjugacy classes of S_4 . We compare four fingerprints:

- (i) the full labelled vector $(a_U(h))_{h \in G}$ in the fixed word order;

- (ii) the five conjugacy-class sums $(\sum_{h \in C_i} a_U(h))_i$;
- (iii) the ordered five-tuple of multisets $(\{a_U(h) : h \in C_i\}_{\text{multi}})_i$;
- (iv) the single unlabelled multiset $\{a_U(h) : h \in G\}_{\text{multi}}$.

These are distinct declared quotients of the data, not interchangeable notions of Patterson equivalence.

Theorem 4.2 (Labelled Patterson separation). *On the 82 rows of Theorem 3.4, the full element-labelled right Patterson vector is injective. The four fingerprints above yield*

	full labelled	class sums	class multisets	unlabelled
# classes	82	20	33	32.

The 24 stored coordinates occupy 17 orbits under inversion.

Proof. For every row the 24 coefficients are computed by exact integer intersection. Equality grouping gives 82 full vectors and, after applying each displayed quotient, 20, 33, 32 classes. The identities of Lemma 4.1 are checked in all $82 \cdot 24 = 1,968$ left-translation cases. Inversion fixes the identity, the six transpositions, and the three double transpositions; the other 14 elements form seven inverse pairs. Thus there are $10 + 7 = 17$ inversion positions. $\square$

Remark 4.3 (Scope). The theorem is a collision-free certificate on one finite family. It is not a phase-retrieval theorem and does not assert that second-order autocorrelation separates arbitrary subsets of S_4 , still less arbitrary finite groups. Higher-order invariants can be complete under hypotheses where second-order data are not [Kak12, RS07].

4.3. The half-size complement ambiguity. For $|U| = |G|/2$, Lemma 4.1 gives $a_{G \setminus U} = a_U$. This universal ambiguity does not produce a second atlas row in the $K12$ family.

Lemma 4.4 (Complement absorption). *For every one of the 50 atlas rows with $|U| = 12$, the complement $G \setminus U$ belongs to the same left tile orbit as U .*

Proof. Every such row has a two-piece left-translate cover $G = g_1 U \sqcup g_2 U$. Multiplication by g_1^{-1} gives $G = U \sqcup gU$, where $g = g_1^{-1} g_2$. Hence $G \setminus U = gU$. The certificate additionally checks the relation directly for all 50 canonical masks. $\square$

The convex catalogue admits a compact separating contact vector [Bab26]. In the nonconvex atlas, the full right-difference labels play the parallel role. The compression profile 82/20/33/32 quantifies exactly how much separation is lost when those labels are forgotten.

5. ALL COVERS AND MULTIPLIER SELECTORS

5.1. Cover orbits and selector fibres. Fix an atlas row U and an exact cover $\mathcal{C}$. Besides the tile stabiliser H_U , two further objects are needed:

$$K_{\mathcal{C}} = \text{Stab}_L(\mathcal{C}) = \{g \in G : g\mathcal{C} = \mathcal{C}\}$$

and the selector-independent lift

$$B_{\mathcal{C}} = \{g \in G : gU \in \mathcal{C}\}.$$

The set of selectors realising this cover is

$$R_U(\mathcal{C}) = \{A \subseteq G : \{aU : a \in A\} = \mathcal{C}\}.$$

Each piece gU has multiplier fibre gH_U , so a selector chooses one element from each such right coset. In particular, selectors encode more than the underlying geometric cover.

Lemma 5.1 (Selector-independent carrier). *For every $A \in R_U(\mathcal{C})$,*

$$B_{\mathcal{C}} = AH_U, \quad K_{\mathcal{C}} = \text{Stab}_L(B_{\mathcal{C}}).$$

Thus $B_{\mathcal{C}}$ depends on U and $\mathcal{C}$, but not on the chosen section A .

Proof. The multipliers producing the piece aU are precisely the elements of the right coset aH_U . Taking the union over $a \in A$ gives $B_{\mathcal{C}} = AH_U$. A group element stabilises the cover if and only if it stabilises its full preimage under the quotient map $g \mapsto gU$, proving the second identity. $\square$

For each row we enumerate every exact cover by two algorithms: the recursive search of Algorithm 3.2 and direct combination testing inside the left tile orbit. Covers are canonicalised by

$$\text{can}_G(\mathcal{C}) = \min_{g \in G} \text{sort}\{m(gV) : V \in \mathcal{C}\}.$$

An independent Burnside computation verifies the orbit counts. Selector orbits over a canonical cover are computed as sections $R_U(\mathcal{C})$ modulo $K_{\mathcal{C}}$, which is equivalent to the global-left action after the cover has been gauge-fixed.

5.2. Complete enumeration.

Theorem 5.2 (Cover and selector census). *The 82 atlas rows have 786 unordered exact covers, forming 86 global-left cover orbits. Seventy-nine rows have one cover orbit. The three exceptions are*

mask suffix	M00007D	M00016D	M001345
# cover orbits	2	3	2.

The 1,148 multiplier selectors form 102 global-left selector orbits. Exactly 90 of those orbits admit a representative that is a subgroup or a one-sided subgroup coset; 2 further orbits admit a genuine double-coset representative but no subgroup/coset representative; the remaining 10 are residual.

TABLE 2. Cover and selector counts by tile size.

$ U $	tile rows	raw covers	cover orbits	selector orbits
6	14	102	18	24
8	18	144	18	21
12	50	540	50	57
total	82	786	86	102

Proof of Theorem 5.2. For each row, the recursive and combination enumerators return the same sorted cover stream. Global-left canonicalisation produces the stated orbit counts, and Burnside’s lemma gives the same result independently. For every cover representative, all selector sections are generated by the right- H_U fibres and quotiented by K_C . The subgroup inventory of S_4 is regenerated from closure and contains exactly 30 subgroups. Testing every selector orbit against that inventory and against all typed one- and two-sided cosets yields the disjoint taxonomy $90 + 2 + 10 = 102$. Every equality is set equality in the 24-element group; no heuristic isomorphism test enters the count. $\square$

The phrase “90 coset selectors” would be too strong. The theorem says that 90 *orbits admit* a subgroup/coset representative; other selectors in the same orbit need not themselves be presented in that form.

5.3. The carrier double coset.

Theorem 5.3 (Typed carrier structure). *For every canonical cover orbit, the selector-independent carrier is one typed double coset:*

$$B_C = K_C a H_U$$

for a suitable $a \in G$. Exactly 85 of the 86 carrier orbits admit a subgroup/coset global-left gauge. The cover attached to *M0030CF* is the unique carrier-level genuine non-coset exception.

Proof. For a fixed piece $aU \in \mathcal{C}$, transitivity of K_C on the pieces of $\mathcal{C}$ gives $B_C = K_C a H_U$. Direct enumeration verifies this equality and the piece transitivity for all 86 cover representatives. The subgroup/coset gauge test is then applied to the carriers themselves, separately from the selector taxonomy, and has the unique stated exception. $\square$

Three normalisers occur in the certificate and must not be conflated: $N_G(H_U)$ for the tile stabiliser, $N_G(K_C)$ for the cover stabiliser, and $N_G(L_A)$ for the left stabiliser $L_A = \{g : gA = A\}$ of a selector. A group element lying in one of these normalisers is not automatically a symmetry of the other typed object.

6. RANK-COLOURED CONTACT COMPLEXES

6.1. Exact intersection data. Every barycentric vertex b_I has rank $|I| \in \{1, 2, 3, 4\}$. A chamber is the simplex on one maximal flag of nonempty subsets of $\{0, 1, 2, 3\}$. For a piece U , let $\mathcal{B}(U)$ be the closed rank-coloured simplicial subcomplex formed by all faces of its chambers. If X, Y are two pieces in one exact cover, define their contact complex by

$$\Omega(X, Y) = \mathcal{B}(X) \cap \mathcal{B}(Y).$$

Because chamber interiors are disjoint, this lies in the boundary shadow. The coned complex retains the universal rank-4 barycentre when present.

For $S \subseteq \{1, 2, 3, 4\}$, let $f_S(\Omega)$ count faces whose vertex-rank set is exactly S . The full coloured flag vector has 15 coordinates. We use the smaller proper-rank marginal

$$\kappa(\Omega) = (f_{\{1\}}(\Omega), f_{\{2\}}(\Omega), f_{\{3\}}(\Omega)).$$

It is important that κ is neither the ordinary f -vector nor the full coloured flag vector. Balanced complexes and flag enumeration provide the classical background [Sta79, Fro12, Whi22].

Two exact contact complexes have the same S_4 type when one simultaneous rank-preserving relabelling of the four tetrahedral vertices maps one labelled complex onto the other. This is a geometric orbit notion, not unrestricted abstract simplicial isomorphism.

6.2. Selected-cover panel.

Theorem 6.1 (Selected contact atlas). *Choose the certified canonical cover for each of the 82 tile rows. The resulting covers contain 188 unordered within-cover piece pairs:*

$$84 \text{ for } |U| = 6, \quad 54 \text{ for } |U| = 8, \quad 50 \text{ for } |U| = 12.$$

Their contact complexes have 28 distinct κ profiles and 84 exact simultaneous- S_4 types. Exactly 11 κ profiles contain more than one exact type.

Proof. The barycentric faces are represented by exact ranked subset flags. For each pair, intersection is performed before any marginal is taken. Acting by all 24 simultaneous label permutations and selecting the least encoded complex gives the exact type. Equality grouping of the 188 records produces the stated counts. Recomputing κ from each full flag vector verifies the proper-rank convention on every record. $\square$

One collision illustrates the information loss. For a particular $\kappa = (2, 1, 2)$ profile with ordinary face data $(6, 9, 4, 0)$, the records split into three different 15-coordinate coloured flag vectors, with multiplicities 4, 18, 4. The marginal sees identical numbers of vertices of each proper rank but forgets how those ranks are incident.

6.3. All covers and a local-to-global obstruction. Now choose one canonical representative of each of the 86 global-left cover orbits. Its complete graph on pieces has one edge per unordered piece pair, labelled by the exact contact type. The 86 panels contain 212 edges and have 82 edgewise signature classes.

Let $E(\mathcal{C})$ denote the unordered piece pairs of a cover. For two covers of the same tile size, *edgewise equivalence* means

$$(E) \quad \exists \pi \forall e \in E(\mathcal{C}) \exists g_e \in S_4 : \Omega'_{\pi(e)} = g_e \Omega_e,$$

where π is induced by one global permutation of the pieces. A *global alignment* requires

$$(G) \quad \exists \pi \exists g \in S_4 \forall e \in E(\mathcal{C}) : \Omega'_{\pi(e)} = g \Omega_e.$$

The same piece permutation occurs in both definitions. Only the order of the geometric quantifiers changes.

Theorem 6.2 (Contact local-to-global obstruction). *There are five unordered pairs of distinct cover-orbit representatives that belong to the same tile row. Exactly two of these pairs satisfy (E): the pairs over M00007D and M001345. None of the five satisfies (G). Consequently edgewise contact*

equivalence does not imply one global tetrahedral alignment on the certified carrier.

Proof. The three multi-cover rows have respectively 2, 3, 2 cover orbits, hence $\binom{2}{2} + \binom{3}{2} + \binom{2}{2} = 5$ pairs. For each pair, the replay enumerates all piece permutations. For (E) it then tests membership in the simultaneous- S_4 orbit edge by edge. For (G) it tests each of the 24 common group elements against every edge. Two pairs pass the first test and all five fail the second. The global test examines 576 cases per pair and 2,880 cases in total. $\square$

The theorem is static. It asserts no hinge motion, collision-free path, continuous deformation, or mechanical realisability.

7. AMBIENT CONGRUENCE, HANDEDNESS, AND TRANSITIVITY

7.1. A subdivision-insensitive corner locus. The boundary triangulation of $|U|_T$ contains vertices introduced only by subdivision of a planar face or a straight edge. To obtain a geometric screen independent of those artefacts, define $C_3(U)$ to be the set of boundary points at which the span of the incident local boundary-plane normals has rank three. Coplanar subdivision does not change this locus.

Lemma 7.1 (Completeness of the mesh-vertex corner screen). *For every chamber union U , each point of $C_3(U)$ is a vertex of the barycentric boundary mesh. Equivalently, computing the normal rank at all boundary mesh vertices finds the complete rank-three locus, whether or not the polyhedral boundary is manifold along every mesh edge.*

Proof. At the relative interior of a boundary triangle there is one local boundary plane, so the normal rank is one. At the relative interior of a mesh edge e , every incident boundary plane contains the direction of e . All its normals therefore lie in the two-dimensional subspace perpendicular to that direction inside the three-dimensional affine space H , so their rank is at most two. This argument does not use manifold edge incidence. Every remaining boundary stratum is a mesh vertex. $\square$

Lemma 7.2 (Corner-locus covariance). *If an affine Euclidean isometry $F : H \rightarrow H$ maps $|U|_T$ onto $|V|_T$, then $F(C_3(U)) = C_3(V)$. If $C_3(U)$ affinely spans H , an isometry is uniquely determined by the images of any affine basis of four locus points.*

Proof. An isometry maps boundary planes to boundary planes and preserves the rank of their normal span, proving covariance. An affine map on a three-dimensional affine space is determined by four affinely independent points. The isometry condition is then an exact equality of Gram matrices. $\square$

7.2. Global rigidity on the eligible carrier. Let

$$\mathcal{E} = \{U \subseteq S_4 : U \neq \emptyset, \quad \Gamma[U] \text{ is connected, } |U| \mid 24\}.$$

The divisibility condition loses no possible congruent cover, because all chambers have the same volume. The exhaustive reduction of Table 1 gives $|\mathcal{E}/S_4| = 2,874$. The global gate constructs $C_3(U)$ for every one of these representatives before making any exact-cover decision. Its load-bearing counts are shown in Table 3.

TABLE 3. Exact all-candidate congruence audit. Pair counts include the diagonal. Class maps are enumerated on canonical S_4 corner classes; all-candidate maps are obtained from them by the recorded transporters.

audit stage	count
eligible canonical candidates	2,874
all unordered candidate pairs	4,131,375
rejected by unequal volume	568,587
rejected by exact corner-distance signature	3,460,040
candidate pairs surviving both screens	102,748
canonical S_4 corner classes	274
canonical class pairs exhausted by both exact lanes	290
exact maps on canonical classes	617
maps derived on all candidate pairs	228,599
cross-class isometric pairs / non- S_4 maps / lane disagreements	0/0/0

Theorem 7.3 (Eligible-carrier congruence rigidity). *For $U, V \in \mathcal{E}$, the polyhedra $|U|_T$ and $|V|_T$ are freely Euclidean congruent if and only if $V = gU$ for some $g \in S_4$.*

Proof. The reverse implication follows from the coordinate isometry Φ_g . Suppose an isometry $F : H \rightarrow H$ maps $|U|_T$ onto $|V|_T$. By Lemma 7.2, it maps $C_3(U)$ isometrically onto $C_3(V)$. The exact audit finds affine rank three for all 2,874 canonical candidate loci. By Lemma 7.1, this is the complete rank-three locus: the 1,809 candidates with nonmanifold mesh-edge incidence require no extra hypothesis, and the independent edge-stratum audit finds zero rank-three nonvertex strata.

Coordinate canonicalisation reduces the candidate loci to 274 exact S_4 classes. Any isometry of two original loci conjugates, by their recorded coordinate transporters, to a map between canonical classes. Exact distance signatures leave the 290 class pairs in Table 3. Weighted distance-graph backtracking and affine-basis Gram reconstruction independently exhaust those pairs and agree on all 617 class-level maps. No two distinct canonical classes are isometric, and every enumerated map is induced by one coordinate permutation Φ_g . The underlying covariance is checked once at the primitive level: 24 chamber bijections, 576 chamber tetrahedra, 2,304 chamber facets, 109,440 facet-incidence comparisons, and 2,304 normal lines imply all $2,874 \cdot 24 = 68,976$ candidate-action cases.

Consequently F and some Φ_g agree on $C_3(U)$. Since that locus affinely spans H , Lemma 7.2 gives $F = \Phi_g$ on all of H . Hence $|V|_T = F(|U|_T) = \Phi_g(|U|_T) = |gU|_T$. The closed Coxeter chambers have pairwise disjoint relative interiors, so their union encoding is injective; therefore $V = gU$. $\square$

Corollary 7.4 (Complete congruent Coxeter-pure atlas). *A connected Coxeter-pure piece covers T by congruent Coxeter-pure copies if and only if it has an exact left-translate cover. Up to free congruence, there are exactly 97 such pieces: 15 convex and 82 nonconvex. The nonconvex classes have*

sizes 6, 8, 12 with multiplicities 14, 18, 50. Thus the nonconvex Coxeter-pure problem of [Bab26] is resolved within the fixed chamber carrier of Section 2.

Proof. In a congruent Coxeter-pure cover, equal volume forces the common chamber count to divide 24, so every member belongs to $\mathcal{E}$. Fixing one member U , Theorem 7.3 writes every other member as gU . Interior disjointness and union T then give an exact left-translate cover. The converse uses the coordinate isometries Φ_g . The counts and the convexity split are Theorem 3.4. $\square$

7.3. Tile congruence and chirality. For every atlas row, $C_3(U)$ spans H . We enumerate its complete weighted distance-graph isomorphisms in two independent ways: backtracking on exact distance profiles and affine-basis Gram reconstruction. Each candidate map is converted, when possible, to the unique coordinate permutation Φ_g and filtered by exact chamber-mask equality. A corner-set symmetry is only a candidate; it need not preserve the full piece.

Theorem 7.5 (Ambient tile classification). *The 82 atlas rows are 82 distinct free Euclidean congruence classes. They split into 159 classes under orientation-preserving congruence. Of the full classes, 77 are chiral and 5 achiral, so*

$$159 = 2 \cdot 77 + 5.$$

For every row, the full Euclidean self-isometry group equals the left tile stabiliser H_U .

Proof. There are 3,321 unordered cross-row pairs and 82 diagonal pairs. Exact distance signatures reject 3,253 cross pairs; the two independent isometry algorithms examine the remaining 68. Across all 3,403 pairs including the diagonal, they produce 539 candidate corner maps. Every candidate is induced by a coordinate permutation. The mask filter leaves no cross-row shape congruence and identifies every self-isometry with H_U . Restricting the coordinate permutations to A_4 yields the oriented split and the stated chirality count. $\square$

This is the finer rowwise statement on the selected nonconvex atlas. Theorem 7.3 supplies the logically prior all-candidate completeness; the exact mask filter here identifies self-isometry groups and the orientation-preserving split.

7.4. Whole partitions. For a cover $\mathcal{C}$, the union of its pieces is the entire tetrahedron T . Any congruence between two whole partitions therefore maps T onto itself, permutes its four extreme vertices, and belongs to the coordinate S_4 . Unlike the tile argument, this reduction requires no corner-locus screen.

Theorem 7.6 (Partition congruence and finite-container transitivity). *The 86 global-left cover orbits are exactly 86 full Euclidean congruence classes of whole partitions. They split into 151 orientation-preserving classes, with 65 chiral and 21 achiral full classes.*

Every one of the 86 partitions is tile-transitive under its full container symmetry group $K_{\mathcal{C}}$. Exactly 69 are tile-transitive under the even subgroup

$K_C \cap A_4$. In the remaining 17 cases the cover stabiliser has type C_4 , and an orientation-reversing symmetry is necessary for piece transitivity.

Proof. An isometry of a whole partition preserves its union T and is therefore a coordinate permutation. Exact cover canonicalisation has already quotiented this full S_4 action, proving the first count. Restriction to A_4 gives the $151 = 2 \cdot 65 + 21$ oriented classes. Direct orbit computation of K_C on the pieces gives one orbit in all 86 cases; repeating with the even subgroup gives one orbit in 69. The stabiliser inventory identifies all 17 failures as type C_4 . $\square$

The term *finite-container tile-transitive* is used to prevent a common over-reading. These are partitions of one tetrahedron; the theorem does not construct or classify periodic tilings of $\mathbb{R}^3$. Moreover, an odd tetrahedral isometry need not be reflection in a plane.

8. TWO EXACT BOUNDARY PRESENTATIONS

The contact complex records how pieces meet one another. A different question asks how much exact metric information is retained by a presentation of one piece's polyhedral boundary. This section records a controlled sensitivity result; it is not used to prove the atlas theorem.

Use the standard simplex of Section 2 and divide squared distances by 2, so the tetrahedron has unit edge length. Boundary edges are classified by their exact incident-plane pair, convex/reflex status, and squared cosine of the dihedral angle. Eight exact classes occur. For each class we sum the unit-normalised squared edge lengths.

The two presentations differ only in subdivision convention.

Atomic presentation.:

Retain the edges of the barycentric boundary triangulation and sum all noncoplanar elementary intervals without collinear merging.

Maximal-reflex presentation.:

Merge connected collinear intervals precisely when the supporting line, the unordered oriented incident-plane pair, and the exact reflex/cosine class agree.

Proposition 8.1 (Presentation sensitivity). *Across the 82 atlas rows, the atomic presentation contains 1,692 counted edges and produces 70 distinct exact profiles. The maximal-reflex presentation contains 1,506 edges and produces 72 profiles. On the full-tetrahedron calibration row, the total unit-normalised squared-edge sum is 3 in the atomic presentation and 6 in the maximal presentation.*

Proof. The boundary triangles are reconstructed from the exact barycentric chamber vertices. Supporting lines, plane normals, reflex signs, squared cosines, and squared lengths are rational or quadratic data compared without floating point. The atomic lane reproduces all 15 rows of the earlier convex appendix; the maximal lane applies the declared merging equivalence independently. Equality grouping gives the displayed edge and profile counts. For the full tetrahedron each original unit edge is split into two half-edges: the atomic contribution is $6 \cdot 2 \cdot (1/2)^2 = 3$, while maximal merging restores the six unit edges. $\square$

Neither presentation is a Dehn invariant. Dehn data are linear in edge length and additive under subdivision, whereas a squared-length sum is non-linear. The numbers 3 and 6 therefore describe two explicit presentation conventions rather than an error and a correction. Classical scissors-congruence and Dehn-invariant results remain separate [Syd65].

9. PRIOR ART AND CLAIM BOUNDARY

9.1. Homometry and reconstruction. Patterson/autocorrelation data and homometric sets have a substantial literature. Rosenblatt and Seymour study structural aspects of homometric sets [RS82]; Mandereau et al. formulate left and right interval contents in a group setting [MGA⁺11]; noncommutative homometry in dihedral groups is treated by Genuys and Popoff [GP18]. Covariograms of nonconvex sets show related reconstruction ambiguity in Euclidean geometry [BBD11]. Higher-order decks or bispectral data can carry information unavailable to a second-order invariant [RS07, Kak12].

Our use of the right Patterson function is therefore classical at the level of definition. The finite result proved here is the exact collision-free statement on the locked 82 rows, together with the measured failure of three label compressions. We do not claim a general reconstruction theorem, nor priority for group-ring homometry. In particular, the 1989 paper of Rosenblatt and Shapiro [RS89] is cited as adjacent semisimple-ring prior art; no theorem from it is imported.

9.2. Coxeter complexes, coloured incidence, and tilings. Coxeter complexes, weak order, and type-selected balanced complexes are classical [Bjö84a, Bjö84b]. Coloured flag enumeration goes back at least to Stanley’s theory of balanced Cohen–Macaulay complexes and its later refinements [Sta79, Fro12, Whi22]. Symbolic chamber-system descriptions of tilings have likewise been developed in a broader setting [DFH95, DH87].

Exact factorisation of finite groups provides a natural algebraic language for multiplier selectors [BGV20]. The typed carrier theorem in Theorem 5.3 is not presented as a new general factorisation theory; it is the complete finite structure of this particular S_4 atlas.

The word “tile-transitive” is also used narrowly. Classical isohedral and monotypic tiling theory concerns much broader ambient spaces and local-to-global questions [GS77, DS04]. Our partitions have one fixed tetrahedral container. We neither produce a space tiler nor classify tetrahedral space-fillers such as the families studied by Sommerville and Goldberg [Som24, Gol74].

9.3. What is and is not classified. The paper classifies connected chamber unions in one fixed 24-chamber A_3 subdivision, with the exact actions and cover notions of Section 2. In a congruent Coxeter-pure cover, every member of the partition remains a union of chambers of that fixed subdivision. Its data include every left-translate cover of each nonconvex tiling row and every multiplier selector in those cover fibres.

It does not classify:

- disconnected chamber unions;
- arbitrary polyhedral pieces or arbitrary tetrahedral dissections;

- congruent copies whose boundaries cut interiors of the fixed chambers;
- right-translate or two-sided group factorisations;
- periodic or face-to-face tilings of $\mathbb{R}^3$;
- Patterson reconstruction outside the displayed finite family;
- contact up to unrestricted abstract complex isomorphism;
- mechanical motion, hinging, or collision-free deformation;
- a presentation-independent squared-edge invariant.

The count $15 + 82$ is already present in [Bab26]. The contribution of the present work is the all-candidate congruence-rigidity theorem that closes the earlier Coxeter-pure problem, together with the canonical nonconvex classification and the exact structural layers that make the 82 rows distinguishable and reviewable.

10. COMPUTATIONAL CERTIFICATION AND REPRODUCIBILITY

10.1. Proof architecture. The computational part of the proof has five layers.

- (1) A from-scratch mask scan constructs the connected and eligible candidate universe, canonicalises left orbits, and runs the exact-cover decision of Algorithm 3.2.
- (2) Before selecting positives, the global congruence gate constructs the rank-three corner locus on all 2,874 eligible candidates, proves its carrier covariance, and exhausts every invariant-surviving isometry in two exact lanes.
- (3) Deterministic builders attach the Patterson, cover, selector, contact, boundary, and ambient-symmetry records to the immutable 82 row identifiers.
- (4) Independent algorithms cross-check the load-bearing quotient counts: source replay versus mask scan, recursive versus combination exact cover, orbit canonicalisation versus Burnside, and distance-graph versus affine-Gram isometry enumeration.
- (5) Validators check schemas, self-hashes, row joins, theorem totals, negative controls, and byte identity of regenerated certificates.

No random seed, numerical tolerance, network service, or proprietary solver is involved. Group, mask, incidence, and flag-complex calculations use integers; ambient coordinates, distances, and plane data use exact rational arithmetic.

10.2. Replay modes. The companion repository provides a short verifier for the pinned theorem certificate and a full reconstruction mode. The short path verifies all displayed totals, row identities, self-hashes, parent pins, complete joins, and the existence of every public artifact cited in the theorem-to-certificate crosswalk. The full path regenerates the connected and eligible census, the all-candidate congruence gate, and the congruent-copy exact-cover decision, then compares all 2,874 candidate records with the canonical certificate. The Patterson, cover, selector, contact, boundary, and ambient-symmetry layers are frozen exact artifacts: the public package validates their provenance, row pins, schemas, joins, and structural identities, but does not regenerate them from their internal production builders. A

fresh-clone gate runs the package in a temporary directory to ensure that no neighbouring repository or private absolute path is required.

The preflight additionally checks:

- deterministic JSON serialisation and SHA-256 manifests;
- absence of private paths and internal routing material;
- a completely green public test suite;
- successful title-named LaTeX compilation with resolved citations and cross-references;
- agreement between every number displayed in the paper and the public theorem certificate.

10.3. Finite proof versus theorem import. The exact enumeration proves statements about the finite carrier once the objects and equivalences are fixed. Euclidean use of the word “convex” additionally imports the connected three-way convexity theorem from [Bab26]. Bibliographic background on Patterson functions, balanced complexes, group factorisation, and tiling terminology is not used as an oracle for any finite count. This distinction is recorded explicitly in the source map and claim ledger shipped with the release-candidate repository.

10.4. Auditability. The complete 82-row catalogue is printed in Section B; larger objects, such as the 212 contact-edge records, are supplied in canonical machine-readable form. Section C maps every theorem to its defining data, replay command, and certificate fields. The public reviewer guide gives a ten-minute reading path and a longer independent replay path. Release status is intentionally separate from mathematical status: the package becomes a publication only after the final external red team and owner-controlled release gate.

11. CONCLUSION

The earlier convex classification exposed a sharp nonconvex remainder of 82 left-translate tiling orbits. The all-candidate rigidity theorem proves that free congruence creates neither additional identifications nor mixed-orbit Coxeter-pure covers, so the left census is the complete answer to the earlier congruent-copy problem. On that foundation, the present work turns the nonconvex remainder into a complete finite atlas: canonical rows, an intrinsic labelled separator, all exact covers and multiplier fibres, exact contact complexes, and full tetrahedral symmetry data. The labelled Patterson vector gives the nonconvex counterpart of the convex contact fingerprint, while the 82/20/33/32 compression profile records why nonabelian coordinate labels are indispensable.

The atlas also displays two independent forms of nonuniqueness. Three tile rows possess more than one cover orbit, and two pairs of those covers agree edgewise without one global contact alignment. At the same time, every whole partition remains tile-transitive inside the tetrahedron. These facts show why tile, cover, selector, local contact, and global symmetry are best treated as separate typed layers rather than compressed into one invariant.

APPENDIX A. SEARCH COMPLETENESS AND INDEPENDENT CHECKS

This appendix makes the finite proof contract explicit. Let $M = 2^{24} - 1$ be the full chamber mask and let $N(U)$ denote the distinct left translates of U .

A.1. Connected-mask scan. For each nonzero integer $m \leq M$, breadth-first search in the induced right-Cayley graph decides connectivity. Each connected mask is emitted once in numeric order. The observed stream has 1,210,648 entries and a pinned stream hash. The eligibility predicate is

$$24 \equiv 0 \pmod{|U|}.$$

Canonicalisation applies all 24 precomputed left-action bit permutations and keeps the minimum. The idempotence checks $\text{can}_L(\text{can}_L(U)) = \text{can}_L(U)$ and $\text{can}_L(gU) = \text{can}_L(U)$ are run throughout the candidate space.

A.2. Exact-cover decision. The decision search memoises remaining masks. If c is the least uncovered chamber, every completion contains one and only one member of $N(U)$ that contains c . This proves the recursion of Algorithm 3.2 complete. For the all-cover census, memoisation is removed from the output layer and the recursion emits every terminal unordered family. A separate combination enumerator tests every $24/|U|$ -subset of $N(U)$; its sorted cover stream agrees exactly with the recursive stream on all 82 positive rows.

A.3. Orbit checks. Tile, cover, and selector canonical keys are computed by explicit minimisation over all 24 group elements. Stabiliser sizes satisfy orbit-stabiliser in every record. Cover and selector orbit totals are independently checked by Burnside averaging. All 30 subgroups of S_4 are generated from closure, not read from a hand-written list; their type histogram is checked against the subgroup inventory in the certificate.

A.4. Geometric checks. Boundary vertices and planes are rational in the standard simplex model. For each of the 2,874 eligible canonical candidates, the gate extracts every boundary triangle, forms primitive normal lines, verifies that no mesh edge supports a rank-three nonvertex stratum, and checks that $C_3(U)$ has affine rank three. This remains valid for the 1,809 candidates whose raw boundary triangulations have nonmanifold edge incidence, by Lemma 7.1.

Covariance under all coordinate permutations is proved at the carrier level rather than sampled mask by mask. Exact audits of chamber bijections, chamber tetrahedra, facet occurrences, equality of facet keys, and primitive normal lines show that boundary extraction and normal rank commute with Φ_g for every one of the 2^{24} masks. This single finite proof implies the 68,976 candidate-action cases used in the canonical reduction.

Complete point-set isometries of the canonical rank-three corner loci are then enumerated by both distance-graph backtracking and affine-Gram reconstruction. The two lanes exhaust the same 290 class pairs, agree on all 617 class-level maps, and find neither a cross-class isometry nor a non- S_4 map. Recorded coordinate transporters derive the corresponding 228,599

maps on the original candidate pairs. On the selected 82 atlas rows, candidate maps are additionally filtered against exact chamber masks to recover self-isometry groups and chirality. Synthetic perturbations, coplanar subdivisions, a corner-set symmetry that fails the mask filter, and a non- S_4 synthetic isometry exercise the rejection paths.

The tests establish completeness inside the declared finite universe. They do not convert the universe into a statement about arbitrary dissections.

APPENDIX B. THE CANONICAL 82-ROW CATALOGUE

The table below is generated from the immutable atlas, cover, fingerprint, and ambient-symmetry records. Here $k = |U|$, $h = |H_U|$, c is the number of raw exact covers, q the number of cover orbits, s the number of selector orbits, and “A/C” denotes achiral/chiral. The mask is the canonical six-digit hexadecimal representative. The complete word lists, Patterson vectors, covers, selectors, contact complexes, and exact hashes are supplied in the machine-readable atlas.

Table 4: Canonical nonconvex atlas. Symbols are defined in the appendix text.

#	mask	k	h	c	q	s	hand	o
1	00006F	6	1	6	1	1	C	2
2	000077	6	1	6	1	1	C	2
3	00007D	6	1	12	2	2	C	2
4	0000CF	6	1	6	1	1	C	2
5	000157	6	1	6	1	1	C	2
6	00016D	6	1	18	3	3	C	2
7	0001CD	6	1	6	1	1	C	2
8	00034D	6	2	3	1	4	A	1
9	001097	6	1	6	1	1	C	2
10	0010C7	6	1	6	1	1	C	2
11	00114D	6	1	6	1	1	C	2
12	0011C5	6	1	6	1	1	C	2
13	001345	6	1	12	2	2	C	2
14	0020BA	6	2	3	1	4	A	1
15	00017F	8	1	8	1	1	C	2
16	0010EF	8	1	8	1	1	C	2
17	00116F	8	1	8	1	1	C	2
18	0014AF	8	1	8	1	1	C	2
19	001F45	8	1	8	1	1	C	2
20	0020DF	8	1	8	1	1	C	2
21	00215F	8	1	8	1	1	C	2
22	00249F	8	1	8	1	1	C	2
23	0030CF	8	2	8	1	4	A	1
24	00314F	8	1	8	1	1	C	2
25	003157	8	1	8	1	1	C	2
26	0031C7	8	1	8	1	1	C	2
27	00348F	8	1	8	1	1	C	2
28	0034CD	8	1	8	1	1	C	2
29	0035C5	8	1	8	1	1	C	2

continued on next page

Table 4 (continued)

#	mask	k	h	c	q	s	hand	o
30	005157	8	1	8	1	1	C	2
31	005353	8	1	8	1	1	C	2
32	007B28	8	1	8	1	1	C	2
33	000FFF	12	4	3	1	3	A	1
34	0017FF	12	1	12	1	1	C	2
35	001FF7	12	1	12	1	1	C	2
36	002BFF	12	1	12	1	1	C	2
37	002FFB	12	1	12	1	1	C	2
38	0033FF	12	1	12	1	1	C	2
39	0055FF	12	1	12	1	1	C	2
40	007FD3	12	1	12	1	1	C	2
41	012B7F	12	1	12	1	1	C	2
42	012FFA	12	1	12	1	1	C	2
43	0417DF	12	1	12	1	1	C	2
44	042BDF	12	1	12	1	1	C	2
45	042DFB	12	1	12	1	1	C	2
46	0433DF	12	1	12	1	1	C	2
47	0447F7	12	2	6	1	1	C	2
48	044DDF	12	2	6	1	2	C	2
49	0455DF	12	1	12	1	1	C	2
50	045DD7	12	1	12	1	1	C	2
51	0471DF	12	1	12	1	1	C	2
52	047DD3	12	1	12	1	1	C	2
53	048EDF	12	1	12	1	1	C	2
54	04D4DF	12	1	12	1	1	C	2
55	04F0DF	12	1	12	1	1	C	2
56	052B5F	12	1	12	1	1	C	2
57	052DFA	12	1	12	1	1	C	2
58	05335F	12	1	12	1	1	C	2
59	05715F	12	1	12	1	1	C	2
60	057DD2	12	1	12	1	1	C	2
61	068E9F	12	1	12	1	1	C	2
62	06D49F	12	1	12	1	1	C	2
63	06F09F	12	1	12	1	1	C	2
64	06F23E	12	2	6	1	1	C	2
65	07AA1F	12	1	12	1	1	C	2
66	07B21F	12	2	6	1	2	C	2
67	07B4BA	12	1	12	1	1	C	2
68	07BA17	12	1	12	1	1	C	2
69	07BE92	12	1	12	1	1	C	2
70	0C5CD7	12	1	12	1	1	C	2
71	0CC3F3	12	2	6	1	1	C	2
72	0D2CFA	12	1	12	1	1	C	2
73	1455D7	12	4	3	1	3	A	1
74	1471D7	12	1	12	1	1	C	2
75	1475D3	12	1	12	1	1	C	2
76	1525FA	12	1	12	1	1	C	2
77	153357	12	1	12	1	1	C	2
78	1575D2	12	1	12	1	1	C	2

continued on next page

Table 4 (continued)

#	mask	k	h	c	q	s	hand	o
79	1725F8	12	1	12	1	1	C	2
80	17A4F8	12	1	12	1	1	C	2
81	1CD1D3	12	2	6	1	1	C	2
82	1CD4C7	12	2	6	1	2	C	2

APPENDIX C. THEOREM-TO-CERTIFICATE CROSSWALK

paper result	principal public artifact	independently checked fields
Theorem 3.4	artifacts/bounded_census.json, artifacts/nonconvex_tile_atlas.json	1,210,648, 67,681, 2,874, 97, 15/82; row stream; 14/18/50 size distribution
Theorem 7.3 and Corollary 7.4	artifacts/global_congruence_gate.json	2,874 spanning loci; zero rank-three edge strata; 290 class-pair comparisons; 617 class maps and 228,599 derived maps; zero cross-class, non- S_4 , and lane-disagreement cases; congruent-cover set = $97 = 15 + 82$
Theorem 4.2 and Lemma 4.4	artifacts/nonconvex_tile_atlas.json	Patterson identities; 1,968 left-action cases; 82/20/33/32 classes; 24/17 labels; 50/50 complement absorption
Theorems 5.2 and 5.3	artifacts/cover_atlas.json	786/86 covers; 1,148/102 selectors; 90/2/10 taxonomy; typed carrier double cosets; unique M0030CF exception
Theorems 6.1 and 6.2	artifacts/contact_obstructions.json	188/28/84/11 selected panel; 212/82 all-cover panel; two edgewise and zero global witnesses
Theorems 7.5 and 7.6	artifacts/ambient_symmetry_atlas.json	82/159, 77/5 tiles; 86/151, 65/21 partitions; 86/69 full/rotation tile transitivity; exact corner-isometry audit
Proposition 8.1	artifacts/boundary_presentations.json	1,692/70 atomic and 1,506/72 maximal-reflex profiles; calibration 3/6

The public aggregate certificate repeats these fields in one stable schema and fails if any displayed number, row digest, or referenced artifact drifts.

DATA AND CODE AVAILABILITY

The release-candidate companion package contains the complete machine-readable atlas, exact standard-library replay code, generated tables, pinned JSON certificates, and a SHA-256 manifest. The package is designed to run without network access or a neighbouring checkout. The public repository URL and archival identifier will be inserted only after the final external red team and owner release gate.

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